

CSCI-338-Computer-Science-Theory
Fall 2025
HW 01

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This assignment is **due on September 09, 2025**. You will need to use Latex to generate a single pdf file and upload it under Assignment 01 on Canvas. There will be a penalty of not using Latex (to finish the assignment). This is not a group-assignment, so you must finish the assignment by yourself. **Figures can be hand drawn and inserted to the pdf.**

1 Problem 01

Prove that $S(n) = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$. Use induction.

Base Case

We check whether the formula holds for the base case $n = 1$.

$$\text{LHS: } S(1) = 1^3 = 1$$

$$\text{RHS: } \frac{1}{4} \cdot 1^2 \cdot (1+1)^2 = \frac{1}{4} \cdot 1 \cdot 4 = 1$$

$$S(1) = \frac{1}{4}(1)^2(1+1)^2$$

Induction Step

We must show that the formula holds for $n+1$ $S(n+1) = 1^3 + 2^3 + \dots + n^3 + (n+1)^3 = \frac{1}{4}(n+1)^2(n+2)^2$

$$S(n+1) = S(n) + (n+1)^3 = \frac{1}{4}n^2(n+1)^2 + (n+1)^3$$

$$S(n+1) = (n+1)^2 \left(\frac{1}{4}n^2 + (n+1) \right)$$

$$\frac{1}{4}n^2 + (n+1) = \frac{1}{4}n^2 + \frac{4(n+1)}{4} = \frac{1}{4}(n^2 + 4n + 4)$$

$$n^2 + 4n + 4 = (n+2)^2$$

$$\text{So: } S(n+1) = (n+1)^2 \cdot \frac{1}{4}(n+2)^2 = \frac{1}{4}(n+1)^2(n+2)^2$$

By mathematical induction, we have proven that

$$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2 \text{ holds for all } n \in \mathbb{N}.$$

2 Problem 02

Find the error in the following proof that $2 = 1$. Consider the equation $a = b$. Multiply both sides by a to obtain $a^2 = ab$. Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$. Now factor each side, $(a+b)(a-b) = b(a-b)$, and divide each side by $(a-b)$ to get $a+b = b$. Finally, let a and b equal 1, which shows that $2 = 1$.

The error in this proof lies in the step where we divide each side by $(a-b)$. This is because the equation we are considering establishes that $a = b$, meaning that $(a-b) = 0$. This proof is invalid because it divides by 0, which would make it undefined in mathematics, and lead to the false conclusion that $2 = 1$.

3 Problem 03

Given a simple planar graph $P = (V, E)$, we have Euler's formula: $|V| + |F| - |E| = 2$, where F is the set of faces of P and E is the set of edges of P . Let $|V| = n$, where V is the set of vertices of P . Prove that $|F|$ is at most $2n$.

$$n + |F| - |E| = 2$$

$$|F| = |E| - n + 2$$

For a simple planar graph with $n \leq 3$: $|E| \leq 3n - 6$

Substitute Into $|F|$: $|F| \leq (3n - 6) - n + 2 = 2n - 4$

Thus: $|F| \leq 2n - 4 < 2n$

This proves the desired inequality for $n \geq 3$.

We now verify the inequality for small values of n :

For $n = 1$: $|F| = 1 \leq 2n = 2$.

For $n = 2$: $|F| = 1 \leq 2n = 4$.

Therefore, $|F| \leq 2n$ holds for all $n \in \mathbb{N}$.

For any simple planar graph with n vertices and $|F|$ faces, we have: $|F| \leq 2n$

4 Problem 04

Prove that in any simple graph there is a path from any vertex of odd degree to some other vertex of odd degree.

Let $G = (V, E)$ be a simple graph, and let $v \in V$ be a vertex with odd degree.

$$\sum_{u \in V} \deg(u) = 2|E|$$

This implies that the number of vertices of odd degree is even.

Thus: if v has odd degree, there must be at least one more vertex $u \in V$ such that $\deg(u)$ is odd.

Let $C \subseteq G$ be the connected component containing v .

Since C is itself a graph we can apply this:

$$\sum_{w \in C} \deg(w) = 2|E_C|$$

Thus: the number of odd-degree vertices in C is also even.

So: v must be connected (by some path) to at least one other vertex of odd degree in its component. Therefore, there exists a path from any odd-degree vertex to some other odd-degree vertex.

5 Problem 05

A fully binary tree T is a tree such that all internal nodes have two children. Prove that a fully binary tree with n internal nodes in total has $2n + 1$ nodes.

Since every node is either internal or a leaf: $t = n + l$

We will prove that: $t = 2n + 1$

which is equivalent to showing that: $l = n + 1$

We prove the claim by induction on n , the number of internal nodes. Base Case: $n = 0$

A fully binary tree with no internal nodes must consist of a single leaf node.

Then: $t = 1 = 2(0) + 1$

Assume that any fully binary tree with n internal nodes has exactly $2n + 1$ total nodes.

That is: $t = 2n + 1$

We want to show that a tree with $n + 1$ internal nodes has $2(n + 1) + 1 = 2n + 3$ nodes.

We start with a fully binary tree with n internal nodes. By the inductive hypothesis, it has $2n + 1$ total nodes. Choose a leaf and make it an internal node by adding two children. This increases the number of internal nodes: $n \rightarrow n + 1$, and the total number of nodes: $2n + 1 \rightarrow 2n + 1 + 2 = 2n + 3$

Thus, the property holds for $n + 1$ as well.

By mathematical induction, a fully binary tree with n internal nodes has exactly: $2n + 1$ total nodes.

6 Problem 06

Given an undirected Tree $T = (V, E)$, the breadth-first search starting at $v \in V$ ($bfs(v)$ for short) finds the shortest path tree starting at vertex $v \in V$. The diameter of the graph G is the longest of all the shortest paths, denoted as $\delta(u, v), u, v \in V$. Suppose you have the following algorithm proposed to compute the diameter of the graph G .

1. Run $bfs(w), w \in V$ and compute the vertex $x \in V$ furthest from w .
2. Run $bfs(x)$ and compute the vertex $y \in V$ furthest from x .
3. Return $\delta(x, y)$ as the diameter of G .

Prove that this algorithm returns the correct diameter of the graph G .

Let $D = \text{diameter}(T) = \max_{u, v \in V} \delta(u, v)$. In a tree, there is a unique simple path between any pair of vertices, so the diameter is realized by a specific pair of nodes (u, v) such that: $\delta(u, v) = D$

We now analyze the algorithm step-by-step.

1.

Since T is a tree, breadth-first search correctly finds shortest paths from w to all other vertices.

Let $x \in V$ be a vertex such that: $\delta(w, x) = \max_{z \in V} \delta(w, z)$

We claim that x must be one of the endpoints of a diametral path. This is because the vertex furthest from any node in a tree must lie at one end of a longest path.

2.

Let $y \in V$ be the vertex such that: $\delta(x, y) = \max_{z \in V} \delta(x, z)$

Because x is already an endpoint of a diametral path, this BFS finds the other endpoint y such that: $\delta(x, y) = D$

3.

The algorithm returns $\delta(x, y)$, which is the length of the longest shortest path in the tree. Thus, the algorithm correctly computes the diameter of the tree.